



Time Series Analysis

Seminar 1: Stochastic processes and stationarity



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Acronyms used in this material

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **GDP**: Gross Domestic Product
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **MA**: Moving Average
- ▣ **PACF**: Partial AutoCorrelation Function

Today's Question and Seminar Structure

- The question of the seminar
 - ▶ **is a series stationary, is it white noise, and how do the ACF and the Ljung–Box test answer these questions?**
- Results to obtain (nothing is handed in)
 - ▶ the answers to the A exercises
 - ▶ a figure and a table for each B exercise
 - ▶ the corrected AI answer of C2
 - ▶ one sentence of interpretation for each exercise
- Notebook for today: [open the seminar notebook in Google Colab](#)
 - ▶ each task names its notebook section
- Exercises, their types and order: the [exercise map](#) (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	mean, autocovariances and ACF of MA(1) and MA(2) processes	Solved, Proposed	A1
A3, A4	a random walk; a trend-stationary series and its first difference	Solved, Proposed	A3
A5, A6	sample ACF, bands, Box–Pierce and Ljung–Box by hand	Solved, Proposed	A5
B1, B2	prices and returns: BET; S&P 500 and EUR/RON	Solved, Proposed	B1
B3, B4	transformations: Romanian real GDP; Romanian HICP inflation	Solved, Proposed	B3
C1, C2	has inflation become more persistent? what is wrong in an AI answer?	Proposed	B1, B4

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
BET, S&P 500	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, not adjusted	1995–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	1996–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped; the last day is 18 September 2026
- Growth and inflation rates in %: log differences over one period or one year ([Prerequisites 13/15](#))
- In the notebook: `b1_returns('bet')`, `sample_acf(x)`, `ljung_box(x, 10)`; no account or key is needed

Prerequisites for Today (1/15): processes and moments

- **Stochastic process** $\{X_t\}$: one random variable for each date t
 - ▶ a time series $x_1, \dots, x_T$ is one observed path of the process
 - ▶ $E[\cdot]$: the expected value, an average over all possible paths at a fixed date

- **Mean and autocovariance**

$$\mu_t = E[X_t], \quad \gamma(t, s) = \text{Cov}(X_t, X_s) = E[(X_t - \mu_t)(X_s - \mu_s)]$$

- ▶ $\gamma(t, s) > 0$: when X_t is above its mean, X_s tends to be above its mean too; $\gamma(t, t) = \text{Var}(X_t)$
 - ▶ correlation: $\text{Corr}(X_t, X_s) = \gamma(t, s) / \sqrt{\text{Var}(X_t) \text{Var}(X_s)}$, between -1 and 1
- **Rules for covariances**, with constants a, b, c
 - ▶ $\text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$; $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$
 - ▶ $\text{Var}(\sum_i a_i X_i) = \sum_i \sum_j a_i a_j \text{Cov}(X_i, X_j)$; for uncorrelated X_i : $\sum_i a_i^2 \text{Var}(X_i)$

Prerequisites for Today (2/15): stationarity

- **Weakly stationary:** finite variance, constant mean, autocovariance that depends only on the lag

$$E[X_t^2] < \infty, \quad E[X_t] = \mu, \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for every } t$$

- ▶ h : the lag, the distance in periods between two dates; $\gamma(0) = \text{Var}(X_t)$
- ▶ a shift in time does not change the mean, the variance or the covariances

- **ACF** (autocorrelation function): the autocovariance in units of the variance

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}, \quad \rho(0) = 1, \quad |\rho(h)| \leq 1, \quad \rho(-h) = \rho(h)$$

- **Strictly stationary:** every joint distribution is unchanged by a shift in time
 - ▶ $(X_{t_1}, \dots, X_{t_k})$ and $(X_{t_1+h}, \dots, X_{t_k+h})$ have the same distribution, for every k , dates $t_1, \dots, t_k$ and shift h
 - ▶ **i.i.d.** (independent and identically distributed): the X_t are independent and all have the same distribution

Prerequisites for Today (3/15): white noise and moving averages

- **White noise** $\varepsilon_t \sim \text{WN}(0, \sigma^2)$: unpredictable shocks

$$E[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_s) = 0 \quad (t \neq s)$$

- ▶ σ^2 : the variance of a shock
- ▶ weak (only uncorrelated), i.i.d. (independent), Gaussian (i.i.d. Normal)

- **MA(q)** (moving average): a weighted sum of the current and the last q shocks

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

- ▶ μ : the mean; $\theta_1, \dots, \theta_q$: the weights of the past shocks; q : the order

- **Autocovariances of an MA(q)**: only the shocks common to X_t and X_{t+h} count

$$\gamma(h) = \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h} \quad (0 \leq h \leq q), \quad \gamma(h) = 0 \quad (h > q)$$

- ▶ with $\theta_0 = 1$; X_t and X_{t+h} share the shocks $\varepsilon_{t+h-q}, \dots, \varepsilon_t$
- ▶ beyond lag q no shock is shared: the ACF cuts off after lag q

Prerequisites for Today (4/15): random walk, AR(1) and trends

- **Random walk:** each value is the previous one plus a shock

$$X_t = X_{t-1} + \varepsilon_t, \quad X_0 = 0 \quad \implies \quad X_t = \varepsilon_1 + \dots + \varepsilon_t$$

- ▶ with drift: $X_t = c + X_{t-1} + \varepsilon_t$; c : the drift, the average change per period
- ▶ every shock stays in all later values

- **AR(1)** (autoregressive): $X_t = \phi X_{t-1} + \varepsilon_t$, with $|\phi| < 1$

- ▶ ϕ : the share of today's value that passes to tomorrow; $\rho(h) = \phi^h$, a geometric decay
- ▶ $\phi = 1$ gives the random walk

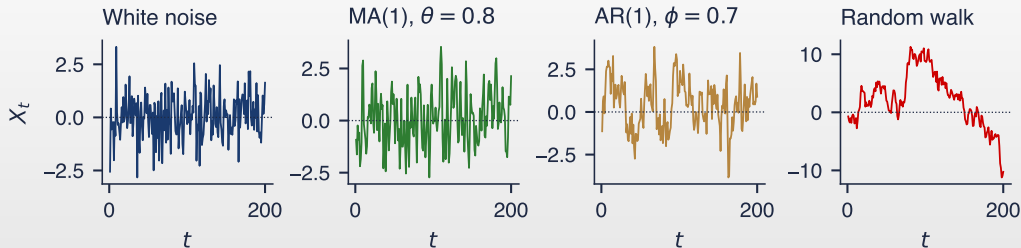
- **Trend-stationary:** $Y_t = a + bt + u_t$, with u_t stationary

- ▶ $a + bt$: a deterministic line; b : its slope, the average change per period

- **Differences**

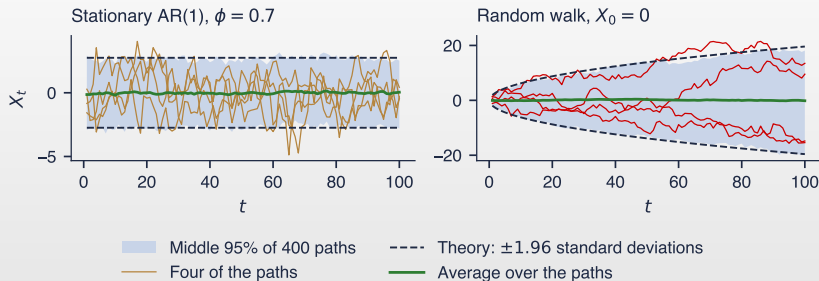
- ▶ $\Delta X_t = X_t - X_{t-1}$; seasonal: $\Delta_s X_t = X_t - X_{t-s}$, s : the number of periods in a year

Prerequisites for Today (5/15): one path of four processes



- ▣ Simulated with the same kind of shocks, $\varepsilon_t \sim N(0, 1)$, $T = 200$
- ▣ From left to right, more and more memory
 - ▶ white noise: no pattern; MA(1): neighbours are similar, the memory lasts one period
 - ▶ AR(1) with $\phi = 0.7$: longer waves, but always back around 0
 - ▶ random walk: no level to return to; the path wanders far from 0

Prerequisites for Today (6/15): many paths at once



- ▣ 400 simulated paths of each process; blue band: the middle 95% of the paths at each date t
- ▣ The mean and the variance at a fixed t are computed across the paths
 - ▶ AR(1): the band has the same width at every t , $\pm 1.96 \sqrt{1/(1 - \phi^2)} = \pm 2.74$: stationary
 - ▶ random walk: the band widens like $\pm 1.96 \sqrt{t}$; the variance grows with t : not stationary

Prerequisites for Today (7/15): the sample ACF and its band

- **Sample autocovariance and sample ACF** (divisor T)

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (x_t - \bar{x})(x_{t+h} - \bar{x}), \quad \hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

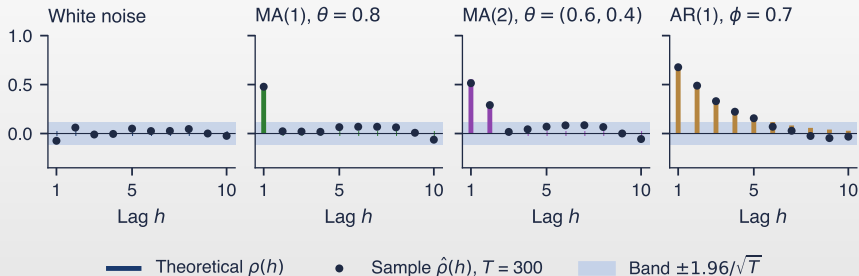
- ▶ $\bar{x} = \frac{1}{T} \sum_t x_t$: the sample mean; the hat marks a value estimated from the data
- ▶ the sample ACF estimates $\rho(h)$ only if the series is stationary

- **Under i.i.d. data:** $\hat{\rho}(h) \approx N(0, 1/T)$, with standard error $1/\sqrt{T}$; the **95% band** is

$$\pm \frac{1.96}{\sqrt{T}}$$

- ▶ 1.96: the 97.5% quantile of the standard Normal distribution
- ▶ a bar outside the band: $\rho(h) \neq 0$ at 5%, for that lag alone
- ▶ with 20 lags, about 1 bar crosses the band by chance

Prerequisites for Today (8/15): theoretical and sample ACF

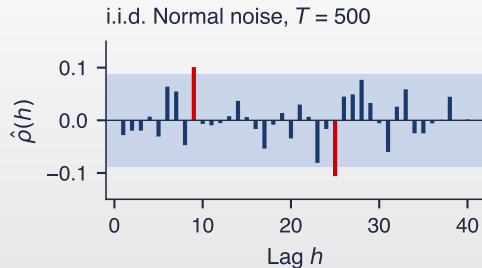


□ Bars: the theoretical $\rho(h)$; dots: $\hat{\rho}(h)$ from one simulated path, $T = 300$

□ Typical patterns

- ▶ white noise: no bar; MA(1): one bar, $\rho(1) = \theta/(1 + \theta^2) = 0.49$; MA(2): two bars, then nothing
- ▶ AR(1): $\rho(h) = 0.7^h$, a geometric decay
- ▶ random walk or trend: a very slow, almost linear decay; seasonal data: peaks at lags $s, 2s, \dots$

Prerequisites for Today (9/15): the band on i.i.d. noise



- One simulated i.i.d. Normal series, $T = 500$: band $\pm 1.96/\sqrt{500} = \pm 0.088$
- 2 of the 40 bars (lags 9 and 25) cross the band
 - ▶ expected: 5% of 40 lags is 2 lags
 - ▶ a single bar just outside the band is weak evidence; the portmanteau tests look at all lags together
- **Significant is not large**
 - ▶ $\hat{\rho}(1)^2$: about the share of the variance of x_t that a linear forecast from x_{t-1} explains
 - ▶ with $T = 5000$, $\hat{\rho}(1) = 0.03$ is outside the band, yet explains only 0.09% of the variance

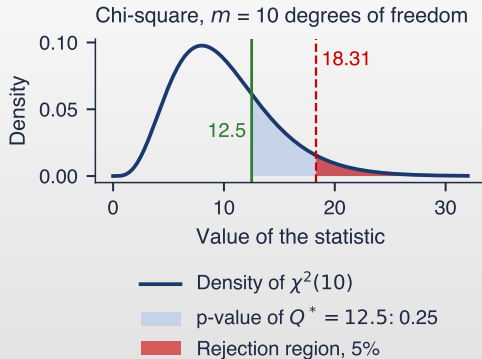
Prerequisites for Today (10/15): portmanteau tests

- H_0 : $\rho(1) = \dots = \rho(m) = 0$, the series is white noise up to lag m
 - ▶ m : the number of lags tested; H_1 : at least one $\rho(h) \neq 0$
- **Box–Pierce** and **Ljung–Box**: sums of squared sample autocorrelations

$$Q(m) = T \sum_{h=1}^m \hat{\rho}(h)^2, \quad Q^*(m) = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}(h)^2}{T-h}$$

- ▶ squares: positive and negative autocorrelations both raise the statistic
- ▶ the factor $(T+2)/(T-h) > 1$ corrects the small-sample behaviour of Q
- ▶ under H_0 both are approximately $\chi^2(m)$, the chi-square distribution with m degrees of freedom
- **Decision**: reject H_0 at 5% if the statistic exceeds $\chi_{0.95}^2(m)$, the 95% quantile of $\chi^2(m)$
 - ▶ 5% critical values: $\chi_{0.95}^2(2) = 5.99$, $\chi_{0.95}^2(3) = 7.81$, $\chi_{0.95}^2(8) = 15.51$, $\chi_{0.95}^2(10) = 18.31$, $\chi_{0.95}^2(12) = 21.03$

Prerequisites for Today (11/15): critical value and p-value

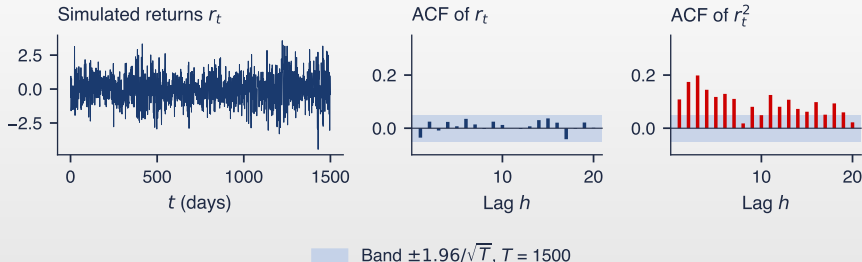


- **p-value:** the probability, under H_0 , of a statistic at least as large as the observed one

$$p = P(\chi^2(m) \geq Q_{\text{obs}}^*)$$

- ▶ Q_{obs}^* : the value computed from the data; p : the area to the right of it
- ▶ reject H_0 at 5% when $p < 0.05$, the same as $Q_{\text{obs}}^* > \chi_{0.95}^2(m)$
- **Chart:** $m = 10$
 - ▶ critical value 18.31: 5% of the area lies to its right (red)
 - ▶ $Q_{\text{obs}}^* = 12.5$ gives $p = 0.25$ (blue area): do not reject white noise
 - ▶ the mean of $\chi^2(m)$ is m : values near m are what H_0 expects

Prerequisites for Today (12/15): volatility clustering: returns and their squares



- **Volatility clustering:** large moves, of either sign, come in groups
 - ▶ simulated daily returns r_t whose variance depends on yesterday's squared return, $T = 1500$
- **Two ACFs of the same series**
 - ▶ r_t : $\hat{\rho}(1) = -0.04$, all bars near the band: the direction of a move is not predictable
 - ▶ r_t^2 : $\hat{\rho}(1) = 0.11$, bars outside the band up to lag 20: the size of a move is predictable
 - ▶ so r_t is uncorrelated, but not independent

Prerequisites for Today (13/15): transformations

- **Logarithm** of a positive series Y_t : $\ln Y_t$
 - ▶ stabilises a variance that grows with the level; a constant growth rate becomes a straight line
- **Growth rates** in % from the log

$$100 \Delta \ln Y_t = 100 (\ln Y_t - \ln Y_{t-1}), \quad 100 \Delta_4 \ln Y_t = 100 (\ln Y_t - \ln Y_{t-4})$$

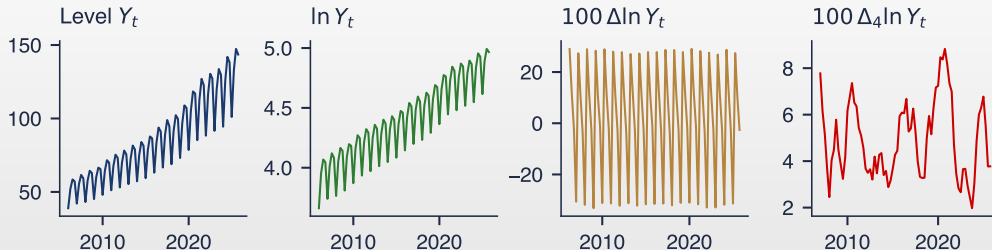
- ▶ Δ : growth over one period; Δ_4 (Δ_{12}): growth over a year for quarterly (monthly) data
- ▶ differences remove trends and random walks; Δ_4 and Δ_{12} also remove the season

- **Box-Cox**: a family of power transformations

$$Y_t^{(\lambda)} = \frac{Y_t^\lambda - 1}{\lambda} \quad (\lambda \neq 0), \quad Y_t^{(0)} = \ln Y_t$$

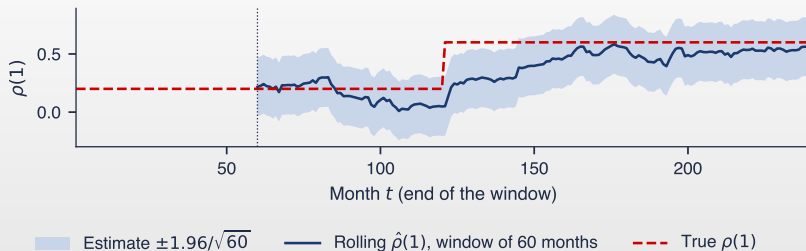
- ▶ λ : the power, chosen so that the variance becomes stable; $\lambda = 1$: no change in shape

Prerequisites for Today (14/15): transformations of a growing seasonal series



- A simulated quarterly series, 2006–2025, growing and seasonal; four views of the same data
- What each transformation does
 - ▶ level: the seasonal swing grows with the level; $\ln Y_t$: the swing has a constant size
 - ▶ $100 \Delta \ln Y_t$: the trend is gone, the season dominates (standard deviation 21.9%)
 - ▶ $100 \Delta_4 \ln Y_t$: trend and season are gone; mean 4.9%, standard deviation 1.7%

Prerequisites for Today (15/15): rolling-window estimates



- ▣ **Rolling window** of length w : the statistic is recomputed on the last w observations at each date t
 - ▶ $\hat{\rho}_t(1)$: the sample $\hat{\rho}(1)$ of $x_{t-w+1}, \dots, x_t$; it shows whether the persistence changes over time
- ▣ **Chart**: a simulated AR(1) whose ϕ jumps from 0.2 to 0.6 at month 120; $w = 60$
 - ▶ the estimate follows the change with a delay of up to w months
 - ▶ standard error about $1/\sqrt{w} = 0.13$; shaded: the estimate ± 1.96 standard errors

A1: an MA(1) process [Solved]

Task

- ▣ $X_t = \varepsilon_t + 0.5 \varepsilon_{t-1}$, with $\varepsilon_t \sim \text{WN}(0, 2)$.
- ▣ 1. Compute the mean $E[X_t]$.
- ▣ 2. Compute $\gamma(0)$, $\gamma(1)$ and $\gamma(2)$.
- ▣ 3. Compute $\rho(1)$ and $\rho(2)$.
- ▣ 4. Say whether the process is weakly stationary, and why.
- ▣ Report: five numbers and one sentence.

Solution

- ▣ 1. $E[X_t] = 0 + 0.5 \cdot 0 = 0$
- ▣ 2. $\gamma(0) = \text{Var}(\varepsilon_t) + 0.25 \text{Var}(\varepsilon_{t-1}) = 2 + 0.5 = 2.50$
- ▣ $\gamma(1) = \text{Cov}(\varepsilon_t + 0.5\varepsilon_{t-1}, \varepsilon_{t+1} + 0.5\varepsilon_t) = 0.5 \cdot 2 = 1.00$;
 $\gamma(2) = 0$ (no common shock)
- ▣ 3. $\rho(1) = 1.00/2.50 = 0.40$; $\rho(2) = 0$
- ▣ 4. Yes: the mean and all $\gamma(h)$ are finite and do not depend on t .

A2: an MA(2) process [Proposed]

Solution discussed in class

Task

- $X_t = 2 + \varepsilon_t + 0.4 \varepsilon_{t-1} - 0.3 \varepsilon_{t-2}$, with $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A1.
- 1. Compute the mean.
- 2. Compute $\gamma(0)$, $\gamma(1)$, $\gamma(2)$ and $\gamma(3)$.
- 3. Compute $\rho(1)$ and $\rho(2)$, and sketch the ACF up to lag 4.
- 4. Say how the constant 2 changes the autocovariances.
- Report: six numbers, the sketch and one sentence.



A3: a random walk [Solved]

Task

- $X_t = X_{t-1} + \varepsilon_t$, $X_0 = 0$, $\varepsilon_t \sim \text{WN}(0, 0.25)$.
- 1. Compute $E[X_{100}]$, $\text{Var}(X_{100})$ and $\text{Var}(X_{120})$.
- 2. Compute $\text{Cov}(X_{100}, X_{120})$ and $\text{Corr}(X_{100}, X_{120})$.
- 3. Say whether X_t and ΔX_t are stationary.
- 4. With a drift $c = 0.1$, compute $E[X_{100}]$.
- Report: six numbers and two sentences.

Solution

- 1. $X_t = \sum_{i=1}^t \varepsilon_i$: $E[X_{100}] = 0$;
 $\text{Var}(X_{100}) = 100 \cdot 0.25 = 25$; $\text{Var}(X_{120}) = 30$
- 2. The common shocks are $\varepsilon_1, \dots, \varepsilon_{100}$: $\text{Cov} = 25$;
 $\text{Corr} = 25 / \sqrt{25 \cdot 30} = \sqrt{100/120} = 0.913$
- 3. X_t : no, its variance grows with t ; $\Delta X_t = \varepsilon_t$: yes, it is white noise.
- 4. $E[X_{100}] = 100 \cdot 0.1 = 10$; the variance does not change.

A4: a trend-stationary series [Proposed]

Solution discussed in class

Task

- $Y_t = 5 + 0.2t + \varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A3.
- 1. Compute $E[Y_t]$ and $\text{Var}(Y_t)$, and say whether Y_t is stationary.
- 2. Write $D_t = \Delta Y_t$ and compute its mean, $\gamma(0)$, $\gamma(1)$ and $\rho(1)$.
- 3. Compare Y_t with a random walk with drift 0.2: what happens to a shock ε_{50} in each?
- Report: five numbers and two sentences.

A5: a sample ACF by hand [Solved]

Task

- ▣ Eight observations: $x = (4, 6, 5, 8, 7, 9, 6, 7)$.
- ▣ 1. Compute $\bar{x}$ and the deviations $x_t - \bar{x}$.
- ▣ 2. Compute $\hat{\gamma}(0)$, $\hat{\gamma}(1)$, $\hat{\gamma}(2)$ with divisor $T = 8$, then $\hat{\rho}(1)$ and $\hat{\rho}(2)$.
- ▣ 3. Compute the 95% band ([Prerequisites 7/15](#)) and compare.
- ▣ 4. Compute the Ljung–Box $Q^*(2)$ and decide at 5%.
- ▣ Report: seven numbers and one sentence.

Solution

- ▣ 1. $\bar{x} = 52/8 = 6.5$; deviations $(-2.5, -0.5, -1.5, 1.5, 0.5, 2.5, -0.5, 0.5)$
- ▣ 2. sum of squares 18: $\hat{\gamma}(0) = 2.250$; products at lag 1 sum to 0.25: $\hat{\gamma}(1) = 0.0312$; at lag 2 to 7.00: $\hat{\gamma}(2) = 0.875$
- ▣ $\hat{\rho}(1) = 0.014$, $\hat{\rho}(2) = 0.389$
- ▣ 3. Band ± 0.693 : both inside
- ▣ 4. $Q^*(2) = 8 \cdot 10 (\hat{\rho}(1)^2/7 + \hat{\rho}(2)^2/6) = 2.02 < 5.99$ ($p = 0.36$): do not reject white noise; with $T = 8$ the test has almost no power.

A6: Box–Pierce or Ljung–Box? [Proposed]

Solution discussed in class

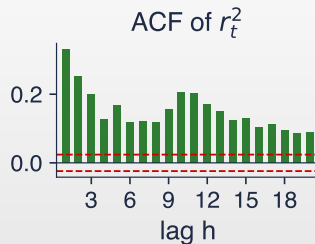
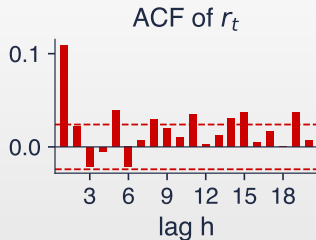
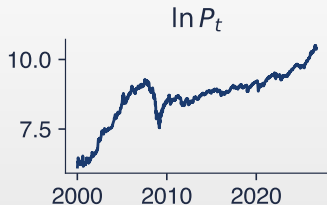
Task

- ▣ A series of $T = 120$ monthly returns has $\hat{\rho}(1) = 0.21$, $\hat{\rho}(2) = -0.08$, $\hat{\rho}(3) = 0.12$. Model: A5.
- ▣ 1. Compute the band and say which autocorrelations lie outside it.
- ▣ 2. Compute $Q(3)$ and $Q^*(3)$.
- ▣ 3. Compare both with $\chi^2_{0.95}(3) = 7.81$ and decide.
- ▣ 4. Explain why the two tests can disagree, and which one you trust here.
- ▣ Report: four numbers and two sentences.

B1: BET prices and returns [Solved]

- **Question:** is the BET log price stationary, and are BET daily returns white noise?
- **Inputs:** BET closes since 2000; $\ln P_t$ and $r_t = 100 \Delta \ln P_t$
- **Tasks**
 1. Compute the sample ACF of $\ln P_t$ at lags 1, 10 and 50.
 2. Draw the ACF of r_t up to lag 20 with the 95% band ([Prerequisites 7/15](#)), and count the bars outside it.
 3. Compute Ljung–Box $Q^*(10)$ for r_t and for r_t^2 , with their p-values.
 4. Interpretation: can yesterday's return be used to forecast today's return?
- **Report:** three ACF values, the chart, two statistics with p-values and two sentences
- **Notebook:** section [B1](#)

B1: solution (1/2) [Solved]

----- $\pm 1.96/\sqrt{T}$

■ sample ACF

B1: solution (2/2) [Solved]

- $\ln P_t$: $\hat{\rho}(1) = 0.999$, $\hat{\rho}(10) = 0.991$, $\hat{\rho}(50) = 0.953$: almost no decay, a random-walk-like level, not stationary
- r_t ($T = 6\,670$, band ± 0.024): 7 of 20 bars outside, the largest $\hat{\rho}(1) = 0.109$; $Q^*(10) = 108.0$, $p < 0.001$
- r_t^2 : $\hat{\rho}(1) = 0.33$; $Q^*(10) = 2440$, $p < 0.001$: much stronger dependence
- Interpretation: yesterday's return explains only $\hat{\rho}(1)^2 = 1.2\%$ of today's variance, too little for a trading rule after costs; the size of yesterday's move does predict today's risk

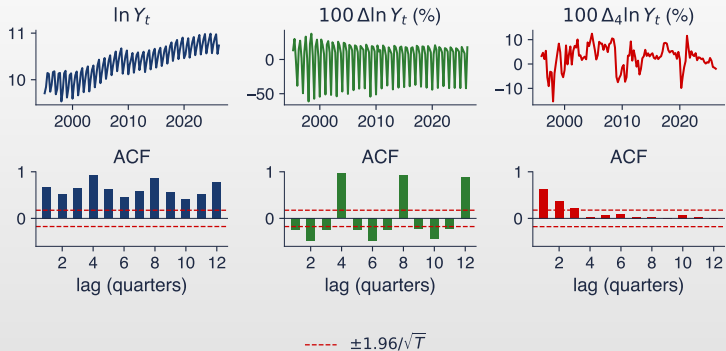
B2: the S&P 500 and EUR/RON [Proposed]

- ▣ **Question:** do the S&P 500 and the EUR/RON rate behave like the BET?
- ▣ **Inputs:** S&P 500 closes since 2000; EUR/RON (BNR reference rate) since July 2005; model: [B1](#)
- ▣ **Tasks**
 1. Repeat the three steps of [B1](#) for both series.
 2. Compare the sign and the size of $\hat{\rho}(1)$ of the returns across BET, S&P 500 and EUR/RON.
 3. Compare $Q^*(10)$ of the squared returns across the three series.
 4. Interpretation: which of the three return series is closest to white noise?
- ▣ **Report:** one table (three series, six numbers each) and two sentences
- ▣ **Notebook:** section [B2](#)

B3: Romanian real GDP [Solved]

- **Question:** which transformation makes Romanian quarterly GDP closest to stationary?
- **Inputs:** real GDP, chain-linked volumes (2010), not seasonally adjusted, Eurostat, from 1995
- **Tasks**
 1. Compute $\ln Y_t$, $100 \Delta \ln Y_t$ and $100 \Delta_4 \ln Y_t$.
 2. Draw the three series and their ACF up to lag 12.
 3. Report $\hat{\rho}(1)$, $\hat{\rho}(4)$ and $Q^*(8)$ for each.
 4. Interpretation: why is $\hat{\rho}(4)$ of the quarterly growth rate close to 1?
- **Report:** a table of nine numbers, the chart and two sentences
- **Notebook:** section [B3](#)

B3: solution (1/2) [Solved]



B3: solution (2/2) [Solved]

	$\hat{\rho}(1)$	$\hat{\rho}(4)$	$Q^*(8)$
$\ln Y_t$	0.68	0.94	495.0
$100 \Delta \ln Y_t$	-0.24	0.96	329.6
$100 \Delta_4 \ln Y_t$	0.63	0.03	74.2

- $\ln Y_t$: trend and season, peaks at lags 4 and 8; $\Delta \ln Y_t$: the season dominates (standard deviation 27.5%); $\Delta_4 \ln Y_t$: mean 2.73%, the ACF dies out after 2–3 quarters
- Interpretation: a q/q rate of unadjusted data repeats the same seasonal swing every year, hence $\hat{\rho}(4) \approx 1$; the annual rate removes the season but reacts late to turning points and overlaps across quarters (positive $\hat{\rho}(1)$)

B4: Romanian inflation [Proposed]

- **Question:** is monthly Romanian inflation since 2005 white noise around its mean?
- **Inputs:** HICP, 2015 = 100, Eurostat, January 2005 to the last month available; model: [B3](#)
- **Tasks**
 1. Compute the monthly inflation $100 \Delta \ln P_t$ and the 12-month inflation $100 \Delta_{12} \ln P_t$.
 2. Draw the ACF of both up to lag 36 and report $\hat{\rho}(1)$, $\hat{\rho}(12)$ and $\hat{\rho}(24)$.
 3. Compute Ljung–Box $Q^*(12)$ for the monthly inflation.
 4. Interpretation: what does the ACF of the monthly inflation say about how fast an inflation shock fades?
- **Report:** six ACF values, one statistic with its p-value, the chart and two sentences
- **Notebook:** section [B4](#)

C1: has Romanian inflation become more persistent? [Proposed]

- **Question:** is monthly inflation more persistent after 2020 than in 2005–2019?
- **Inputs:** monthly HICP inflation of [B4](#), split in January 2020; models: [B1](#), [B4](#)
- **Tasks**
 1. Compute $\hat{\rho}(1)$ and $Q^*(12)$ in each subperiod, with the band of each.
 2. Draw $\hat{\rho}(1)$ on rolling windows of 60 months.
 3. List the events of 2020–2024 that could change the persistence.
 4. Interpretation: why is the difference of two sample autocorrelations hard to judge with 80 observations?
- **Report:** a table, the chart and a plan for a project
- **Notebook:** section [C1](#)

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to interpret the S&P 500 daily data since 2000 ($T = 6\,714$ returns). The answer:
- (a) The ACF of the log price at lag 1 is 0.999, so the log price is a stationary process with a very long memory.
- (b) The return autocorrelation at lag 1 is -0.099 , outside the band of 0.024, so a trader can forecast most of tomorrow's return.
- (c) Ljung-Box $Q(10)$ of the squared returns is 6133, so the returns themselves are autocorrelated.
- (d) A weakly stationary process is always strictly stationary, since its mean and variance are constant.
- (e) Differencing a series that is already stationary does no harm: white noise stays white noise.
- (f) A random walk has $\text{Var}(X(t)) = t * \sigma^2$, so it is not stationary.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ Autocovariances of linear processes follow from the covariance rules: only common shocks count
- ▣ A random walk has a growing variance; its difference is white noise; a trend-stationary series becomes an over-differenced MA(1)
- ▣ Look at the level, the changes and the squares: prices wander, returns are almost uncorrelated, squares are strongly correlated
- ▣ For macro data, choose the transformation first ($\log$, Δ , Δ_4 , Δ_{12}), then read the ACF
- ▣ An AI answer is a draft: check what is stationary, what “significant” means, and whether squares or levels were tested

After the seminar

- Lecture 1 develops each topic of today: stationarity, white noise and the random walk, Wold, ergodicity, ACF and PACF, the portmanteau tests, transformations
- C1 can grow into a team project: inflation persistence in Romania and in the region (Hungary, Poland, Czechia)
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 1–2; [Hyndman and Athanasopoulos \(2021\)](#), Ch. 2–3; [Brockwell and Davis \(2016\)](#), Ch. 1

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